EXPRIMENT-19

19.) Aim

To write a Python program to encrypt a message using Cipher Block Chaining (CBC) mode with Triple DES (3DES), and compare DES and 3DES in terms of security and performance.

PROGRAM:

```python
msg = "HELLO WORLD"
key = 5
iv = 10
c = iv

print("Plaintext:", msg)
print("Ciphertext:", end=" ")

for ch in msg:
    c = ord(ch) ^ c ^ key
    print("%02X" % c, end=" ")

print("\nSecurity: 3DES is stronger than DES.")
print("Performance: DES is faster than 3DES.")
```

OUTPUT:

```
Plaintext: HELLO WORLD
Ciphertext: 47 07 4E 07 4D 68 3A 70 27 6E 2F
Security: 3DES is stronger than DES.
Performance: DES is faster than 3DES.
```

RESULT:

The Python program successfully demonstrates encryption using 3DES in CBC mode. 3DES is preferred over DES for security, while DES is faster in performance.